

NOOR FATIMA

Full-Stack Development. Lahore, Pakistan

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PROFESSIONAL SUMMARY

BS Software Engineering student at PUCIT Lahore (CGPA 3.5/4.00) with hands-on experience building full-stack web applications, REST APIs, and data-driven systems. Proficient in Python, C++ and C#, with strong foundations in OOP, clean architecture, and database design. Built 10+ end-to-end projects spanning ASP.NET Core MVC, SQL Server, ML pipelines, and Computer Vision. Teaching Assistant for 200+ students with proven communication, analytical, and problem-solving skills. Seeking Software Engineering, Full-Stack, or ML internship opportunities, remote or Lahore-based. Available from July 2026.

EDUCATION

BS Software Engineering | CGPA: 3.5 / 4.00 Aug 2023 – May 2027

Punjab University College of Information Technology (PUCIT) · Lahore

Relevant Coursework: Machine Learning, Computer Vision, Data Structures & Algorithms, Database Systems, Software Engineering, Web Development, OOP, Operating Systems, Linear Algebra, Statistics

PROFESSIONAL EXPERIENCE

Teaching Assistant — Software Engineering & OOP 2025 – 2026

PUCIT · Lahore, Pakistan

- ▶ Mentored 200+ students per semester in OOP, software architecture, system design, and debugging, which strengthened technical communication and explanation skills across the department
- ▶ Conducted structured code reviews on 50+ projects, enforcing SOLID principles, clean architecture, and analytical problem decomposition
- ▶ Designed and delivered lab sessions on testing, clean architecture, and software best practices for cohorts of 200+ students
- ▶ Worked with faculty to plan curriculum content, improving lab engagement across two consecutive semesters

ENGINEERING PROJECTS

Virtual Paint — Real-Time CV Application

2026 · [GitHub](#)

Python · OpenCV · NumPy · HSV Color Detection · Real-Time Image Processing

- ▶ Built a real-time virtual painting application using a webcam and OpenCV; users draw on screen by pointing colored objects at the camera with no mouse or touchscreen required
- ▶ Implemented HSV color space detection to isolate multiple colors simultaneously under varying lighting conditions; applied contour detection to extract brush tip position frame by frame
- ▶ Built full UI controls including multi-color support, adjustable brush sizes, eraser tool, canvas clear, and PNG export — all rendered live on the video feed

Feline Gallery — MVC Web Application

2026 · [GitHub](#)

C# · ASP.NET Core MVC · Razor Pages · HTML5 · CSS3 · Bootstrap · SQL Server · Entity Framework Core

- ▶ Built a full-stack MVC web application with layered architecture: ASP.NET Core controllers handle routing and business logic, Razor Views render dynamic content, and EF Core manages gallery data in SQL Server, mirroring enterprise application structure
- ▶ Implemented responsive frontend with Bootstrap and structured navigation across gallery sections; applied separation of concerns throughout with each layer having a clearly defined responsibility
- ▶ Designed dynamic content management for images, descriptions, and categories, keeping the UI fully data-driven with no hardcoded content

Spam Email Classifier — NLP Pipeline

2026 · [Live Demo](#)

Python · NLTK · TF-IDF · Scikit-learn · Naive Bayes · Logistic Regression · SVM · Streamlit

- ▶ Built an NLP pipeline covering text preprocessing, TF-IDF vectorization, and classifier training, achieving 97%+ accuracy on a real-world email dataset
- ▶ Trained and benchmarked Naive Bayes, Logistic Regression, and SVM models; evaluated with precision, recall, and F1-score to pick the best performer
- ▶ Iterated on model evaluation with attention to detail across multiple classifier configurations; deployed as a live Streamlit web app

Car Price Predictor — Regression Pipeline

2026 · [Live Demo](#)

Python · Pandas · Scikit-learn · Linear Regression · OneHotEncoding · Streamlit

- ▶ Built an end-to-end regression pipeline on 10,000+ real used car listings from Quikr India, handling messy real-world data — garbage year values, mixed-format prices, inconsistent kms
- ▶ Performed full EDA with interpretations revealing brand as the strongest price predictor over year or kms driven; removed outliers above 60 lac that skewed model performance
- ▶ Wrapped OneHotEncoding and Linear Regression in a single Scikit-learn Pipeline to prevent data leakage; benchmarked across 1000 random train/test splits to find the most stable model
- ▶ Deployed as a live Streamlit web app with dynamic dropdowns filtered by company selection

ML Algorithms from Scratch

2026 · [GitHub](#)

Python · NumPy · Pandas · Matplotlib · Jupyter Notebook

- ▶ Implemented Decision Tree (ID3), K-Means Clustering, Linear Regression, and Logistic Regression from first principles with no Scikit-learn shortcuts
- ▶ Manually computed entropy, information gain, gradient descent, and sigmoid functions; evaluated each model with full metric suites

RPG Adventure — Text-Based Game

2025 · [GitHub](#)

C++ · OOP · File I/O · Game Logic

- ▶ Developed a text-based RPG in C++ applying OOP principles including classes, inheritance, and polymorphism to model players, enemies, and game state; demonstrates strong C++ programming and analytical design ability
- ▶ Implemented inventory management, combat mechanics, and branching narrative logic with clean modular design and file-based save/load functionality

MNIST Digit Classification — Handcrafted Feature

2026 · [GitHub](#)

Python · NumPy · Scikit-learn · Matplotlib · Feature Engineering

- ▶ Classified MNIST handwritten digits using manually engineered pixel-based features rather than deep learning, demonstrating a strong understanding of classical ML pipelines
- ▶ Extracted handcrafted features (pixel intensities, histogram bins, spatial statistics); trained and evaluated classifiers with precision, recall, and confusion matrix analysis

TECHNICAL SKILLS

Languages Python, C++, C#, SQL

Frameworks ASP.NET Core, .NET MVC, Blazor, Bootstrap, Web APIs, REST APIs, Clean Architecture

AI / ML Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, NLTK, TF-IDF, Streamlit

Databases SQL Server, PostgreSQL, MySQL, Oracle, Entity Framework Core

Tools n8n, Google APIs, Git & GitHub, Postman, Jupyter Notebook, Google Colab, VS Code

Concepts OOP, MVC, SOLID, SDLC, Data Structures & Algorithms, Agile

Soft Skills Teamwork, Communication, Adaptability, Problem-solving, Mentoring, Attention to Detail

HONORS & ACTIVITIES

- ▶ Top Campus Ambassador, LoopLab & LoopVerse (2025)
- ▶ Participant, COSMOCON'25 Web Development Hackathon (2025)
- ▶ Participant, Code Bees 4.0 Competitive Programming Contest (2025)